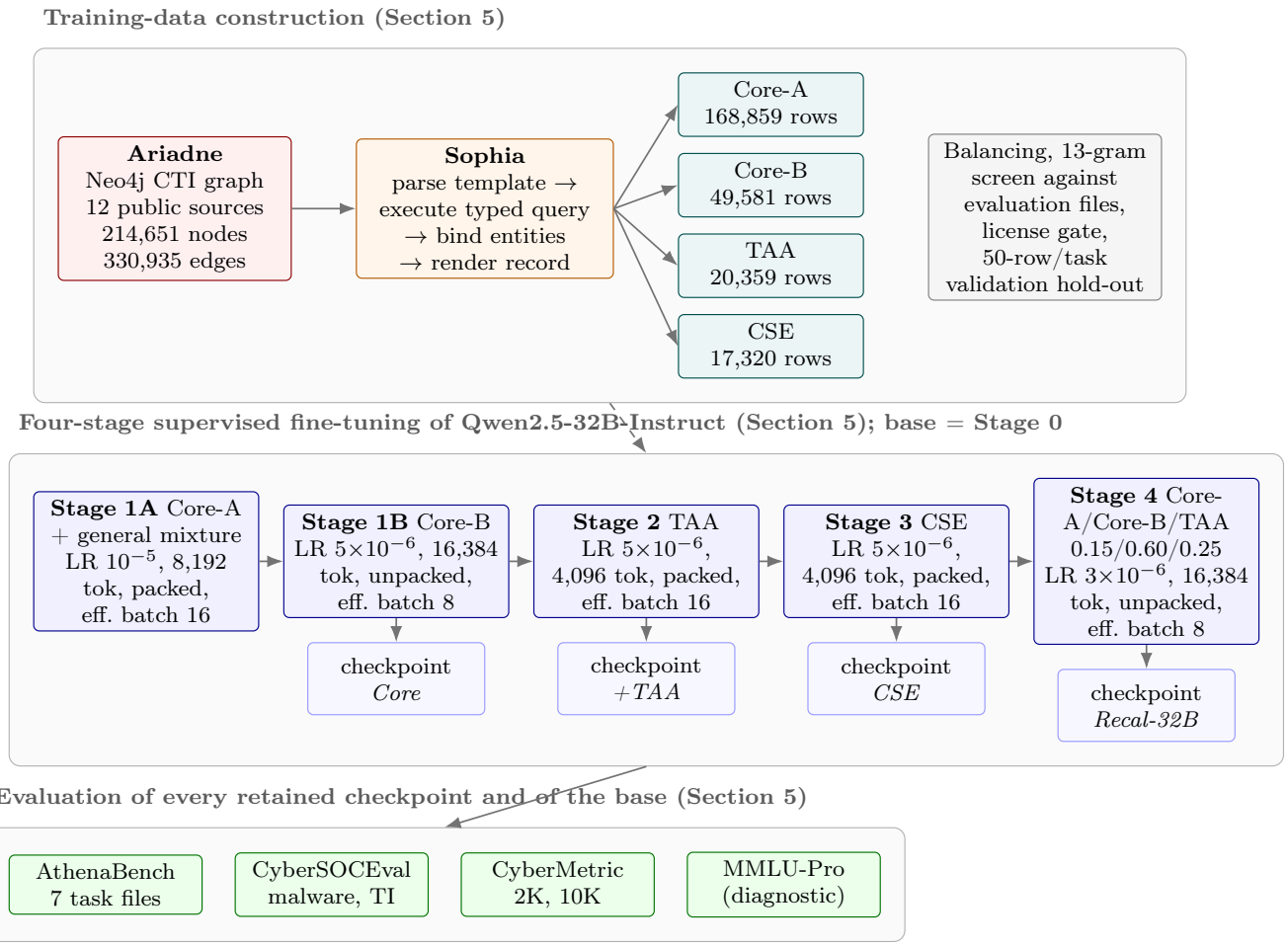

Training-data construction (Section 5)
Ariadne
Neo4j CTI graph
12 public sources
214,651 nodes
330,935 edges
Sophia
parse template →
execute typed query
→ bind entities
→ render record
Core-A
168,859 rows
Core-B
49,581 rows
TAA
20,359 rows
CSE
17,320 rows
Balancing, 13-gram
screen against
evaluation files,
license gate,
50-row/task
validation hold-out
Four-stage supervised fine-tuning of Qwen2.5-32B-Instruct (Section 5); base = Stage 0
Stage 1A Core-A
+ general mixture
LR 10−5, 8,192
tok, packed,
eff. batch 16
Stage 1B Core-B
LR 5×10−6, 16,384
tok, unpacked,
eff. batch 8
checkpoint
Core
Stage 2 TAA
LR 5×10−6,
4,096 tok, packed,
eff. batch 16
checkpoint
+TAA
Stage 3 CSE
LR 5×10−6,
4,096 tok, packed,
eff. batch 16
checkpoint
CSE
Stage 4 Core-
A/Core-B/TAA
0.15/0.60/0.25
LR 3×10−6, 16,384
tok, unpacked,
eff. batch 8
checkpoint
Recal-32B
Evaluation of every retained checkpoint and of the base (Section 5)
AthenaBench
7 task files
CyberSOCEval
malware, TI
CyberMetric
2K, 10K
MMLU-Pro
(diagnostic)